

LongMedBench: Benchmarking Medical Agents for Long-Horizon Clinical Decision-Making

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Abstract. In this work, we introduce LongMedBench, a real-world EHR-based benchmark for long-horizon clinical decision-making. Prior evaluations of LLM-based medical agents have largely emphasized short-context knowledge QA and tool use. However, real-world medical care is inherently longitudinal, and clinicians must aggregate evidence across repeated visits, tests, and evolving treatments. Therefore, long-horizon interaction is essential for realistic assessment. LongMedBench is constructed via a reproducible pipeline that integrates MIMIC-IV admission records and clinical notes into time-series event streams and long-context memory datasets, enabling long-horizon, multi-session interactions between agents and a clinical environment. It comprises 335 patients, with 19.72 inpatient visits per patient on average and 44.91 medical events per visit. Guided by the long-horizon decision process, we propose an evaluation taxonomy with three suites: fact-based QA, temporal reasoning, and long-horizon decision-making. This taxonomy measures how agents understand and leverage historical patient information over extended horizons. Our experiments show that while recent LLMs can make good use of explicit timestamps, they have challenges in implicit time inference; The RAG and agent memory system can improve the performance of information retrieval tasks, but the performance of decision-making tasks is highly dependent on the model’s immediate context.

Keywords: Computer-Aided Diagnosis · Medical Agents · EHR.

1 Introduction

In clinical practice, the diagnostic process is inherently longitudinal and time-dependent. Clinicians do not merely react to isolated symptoms; instead, they must synthesize evidence across multiple visits, diagnostic tests, and evolving

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Table 1: Comparison of Medical Agent Benchmarks.

Benchmark	Long. ^a	Ctx. ^b	EHR. ^c	Dec. ^d	Temp. ^e	Focus
MedAgentBench[6]	×	×	✓	✓	×	EHR Tool Integration
AgentClinic[17]	×	×	×	✓	×	Simulated Interaction
DiagBench[15]	×	×	✓	✓	×	Diagnostic Trajectory
MedBench v4[3]	×	✓	✓	×	×	Medical Knowledge QA
ReflecTool[10]	×	×	×	✓	×	Reflective Tool-Use
EHRSQL[9]	×	×	✓	×	×	Relational Querying
LongMedBench	✓	✓	✓	✓	✓	Long-horizon Reasoning

^a*Longitudinal*: Reasoning across multiple discrete clinical visits. ^b*Context*: Whether the benchmark environment contains multi-turn context. ^c*EHR*: The dataset built upon real-world EHR. ^d*Decision-making*: Evaluating proactive clinical planning beyond static retrieval or SQL-based querying. ^e*Temporal Sensitivity*: Evaluating understanding of clinical timeline and urgency.

treatment responses over years [4]. This capacity of **long-horizon clinical reasoning** is fundamental to high-quality care. As Large Language Models (LLMs) transition to autonomous medical agents, their ability to navigate complex trajectories in real-world Electronic Health Records (EHR) [7] has become the critical benchmark for clinical readiness.

Although general-purpose benchmarks have emerged to evaluate long-horizon or multi-turn interactions [19, 18, 8, 20, 5, 3], these benchmarks primarily emphasize **explicit reasoning based on information retrieval**—the ability to locate specific facts within a long context [11]. They test a model’s capacity to find a “needle in a haystack”, but overlook the temporal dynamics that are essential to clinical reasoning. In real-world medical scenarios, the challenge extends beyond retrieving isolated facts to understanding how a patient’s state evolves over dozens of visits and how this temporal progression informs treatment strategies.

In parallel with these general limitations, current medical evaluation paradigms also fall short of this requirement, as shown in Table 1. Despite introducing simulated interactive environments, recent frameworks [15, 6, 17, 3, 10, 9] are **constrained by limited context windows and session counts**, emphasizing the agent’s tool-call abilities and immediate prediction rather than understanding of complete clinical trajectories. Consequently, they fail to assess how models utilize extensive medical history for future clinical decisions.

To address these limitations, we introduce LongMedBench, a MIMIC-IV[7]-based benchmark that overcomes constrained context lengths and simple factual recall by curating extensive longitudinal trajectories and designing reasoning tasks that challenge agent’s temporal sensitivity. By converting 335 patient records into longitudinal event streams (averaging 19.72 visits per patient), we construct a temporally dense environment and a framework explicitly targeting long-horizon clinical reasoning. Our contributions are: (1) **A tri-level memory dataset architecture** for different granularity history evaluation. (2) **A progressive evaluation taxonomy** that spans three hierarchical tasks: **factual QA** based on timestamp or relative positioning, targeting the fact retrieval limitation of general benchmarks; **temporal reasoning** for multi-visit and event-level ordering, addressing the lack of time-sensitive evaluation in existing medical

frameworks; and **long-horizon decision-making** which directly challenges the agent’s ability to navigate extensive histories and autonomously plan next-step clinical actions. (3) **Experimental findings**: we showed that while state-of-the-art LLMs can exploit explicit timestamps, they struggle with the implicit temporal reasoning required for visit-level understanding. Although RAG and memory systems improve fact retrieval performance, decision-making accuracy remains highly dependent on immediate context, highlighting a profound limitation in reasoning over long-term clinical trajectories.

2 Methodology

2.1 EHR Data Processing Pipeline

LongMedBench is built on MIMIC-IV[7], a public database that contains medical records for $> 100,000$ patients. To convert static EHRs into an interactive agent environment, we design a three-stage pipeline (Figure 1).

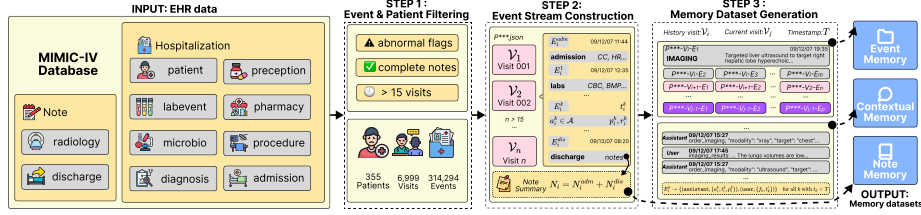


Fig. 1: Data processing pipeline for LongMedBench

Event & Patient Filtering To reduce redundancy while preserving clinical signals, we retain only abnormal lab results and patients with complete admission/discharge records. Filtering for patients with ≥ 15 hospitalizations yields 355 patients and 6,999 visits. With an average of 19.72 visits per patient (median=18.00, SD=5.72), this dense multi-session structure rigorously evaluates the agent’s ability in cross-session memorization and temporal reasoning.

Event Stream Construction A complete event stream will be constructed for each patient $\mathcal{P}$. The event stream $\mathcal{S}$ contains a patient’s visit records $\mathcal{S} := \{\mathcal{V}_1, \mathcal{V}_2, \dots, \mathcal{V}_n\}$, where $\mathcal{V}_i$ is the i_{th} visit of the patient and $n \geq 15$. On average, each visit $\mathcal{V}_i$ contains 44.91 medical events (median=25.00, SD=70.32).

A visit $\mathcal{V}_i$ begins with an admission event E_i^{adm} , followed by a series of specific medical events, and ends with a discharge event E_i^{dis} . The k_{th} medical event in $\mathcal{V}_i$ is denoted as $E_i^k = \{a_i^k, t_i^k, p_i^k, o_i^k\}$, where a_i^k is an action within a typical clinic action space $\mathcal{A}$, which includes *imaging*, *lab tests*, *medication*, etc. t_i^k is the event timestamp, and p_i^k is the corresponding action argument, such as the modality (CT, X-ray, MRI) of an imaging event. o_i^k is the event result or clinical

observation, *like the radiology report in an imaging event*. $\mathcal{V}_i$ can therefore be expressed as a union of sequenced events: $\mathcal{V}_i = \{E_i^{adm}, E_i^1, E_i^2, \dots, E_i^{dis}\}$.

In addition to the structured hospitalization data, each visit also includes two types of notes: one is radiology notes, which are parsed as imaging events; the other is discharge notes, which are summaries N_i from visits $\mathcal{V}_i$, and can be parsed into admission info N_i^{adm} and discharge info N_i^{dis} according to logic.

Memory Dataset Generation To evaluate how agents utilize long-context information, we design three memory modules with progressively finer granularity. From the beginning history visit $\mathcal{V}_i$, current visit $\mathcal{V}_j$ and reasoning timestamp T , the agent’s memory access is strictly bounded to prevent event leakage.

1. Note Memory: $\mathcal{M}_N^{(i,j)} = \{N_i, N_{i+1}, \dots, N_{j-1}\}$, which contains visit-level summaries from preceding $\mathcal{V}_i$ to $\mathcal{V}_{j-1}$. The summary is high-level compressed and emphasizing global trajectory rather than detailed event recall.

2. Event Memory: $\mathcal{M}_E^{(i,j)} = \bigcup_{k=i}^{j-1} \mathcal{V}_k$, which contains all clinical events from $\mathcal{V}_i$ to $\mathcal{V}_{j-1}$. Events are flattened into a single chronological sequence, discarding visit-level boundaries, and forming a unified patient trajectory.

3. Contextual Memory: $\mathcal{M}_C^{(j,T)} = \{(r_0, m_0), (r_1, m_1), \dots, (r_P, m_P)\}$, where r_k and m_k denote the dialog role and message content. It is constructed by rewriting the event stream of current $\mathcal{V}_j$ into an LLM-style trajectory. Only events E_j^k with $t_j^k < T$ are retained. Each E_j^k is transformed into a action-feedback pair using LLM: $\{(\text{assistant}, \{a_j^k, t_j^k, p_j^k\}), (\text{user}, \{f_j^k, t_j^{k'}\})\}$, where f_j^k and $t_j^{k'}$ denote simulated clinic feedback inferred from o_j^k and corresponding timestamp. This memory simulates the instant interaction status for a medical agent.

2.2 Benchmark Question Generation

Based on the memory modules $\{\mathcal{M}_N^{(i,j)}, \mathcal{M}_E^{(i,j)}, \mathcal{M}_C^{(j,T)}\}$, we construct three progressively challenging task families, from *factual QA*, *temporal reasoning*, to *long-horizon decision making*, reflecting increasingly global memory dependency.

Factual QA This task evaluates precise retrieval and temporal alignment within the agent’s event memory. From visit $\mathcal{V}_i$, a ground truth event $E_i^k \in \mathcal{V}_i$ is randomly sampled, and is unified as questions using the following two formats: **Explicit** tasks include t_i^k in the query text, evaluating direct retrieval capacity; while **Relative** questions provide only relative temporal relationships without timestamp (*e.g. the medicine name in the 2nd medication event of $\mathcal{V}_i$*). The questions require the agent to recover full factual details p_i^k from a redundant memory window $\mathcal{M}_E^{(i-m, i+m+1)}$ provided, where $m \geq 0$ is a hyperparameter to adjust the window size. We place $\mathcal{V}_i$ in the middle of the window to evaluate retrieval from non-recency-favored regions of long-context memory [11].

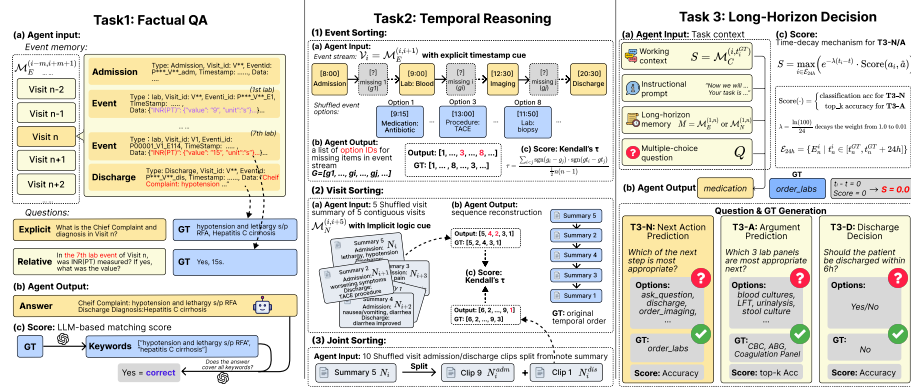


Fig. 2: Example of the three evaluation tasks in LongMedBench.

Temporal Reasoning This task evaluates the agent’s ability to reconstruct chronological order at different memory granularities. The evaluation score is measured by Kendall’s τ . Given a target event stream $\mathcal{S}$ or its visit $\mathcal{V}_i$, we design three sub-tasks using the original event streams as ground truth, each targeting at a distinct level of temporal reasoning. **Visit Cloze** tests event-level reasoning: several intervention events $E_i^k \in \mathcal{V}_i$ with timestamps preserved are masked and provided to the agent as a shuffled list. The agent must insert each event back to its original position. **Visit Sorting** evaluates visit-level ordering: five shuffled visit summaries $N \in \mathcal{M}_N^{(i, i+5)}$ must be ordered solely by clinical progression cues with their timestamps removed. **Joint Sorting** assesses integration: five visits are split into ten admission/discharge clips. The agent must correctly pair the clips from one visit and order them in time sequence.

Long-Horizon Decision Making This task evaluates agent’s next-step planning under long-term historical context, a more comprehensive evaluation. From $\mathcal{V}_n$, we sample a ground-truth event E_i^{GT} , and define the agent’s contextual memory as its working context S . The long-horizon note or event memory is also provided as a reference M . When generating question Q , we randomly select one of the following three sub-task formats: **Next Action Prediction (T3-N)** requires the agent to predict a_i^{GT} , **Argument Prediction (T3-A)** evaluates inference for p_i^{GT} , and **Discharge Decision (T3-D)** queries if the patient can be discharged within 6h. For two *prediction* tasks, as shown in Figure 2, a time-decay scoring mechanism is applied to recognize actions $\mathcal{E}_{24h}$, events that occur within 24h after t_i^{GT} . Unlike prior tasks, this task requires integrating immediate state S with long-term memory M to produce clinically coherent decisions. It evaluates true long-horizon reasoning rather than isolated memory access.

3 Experiments and Results

3.1 Experiment Environments

To verify the robustness of medical agents in long-term clinical decision, we selected representative long-context LLMs as the agent’s backbone, including the closed-source frontier model *gpt-5-mini*[13], the lightweight one *qwen-turbo*[16], and the open-source high-performance one *deepseek-v3.2*[2]. All LLMs are deployed with default configurations.

Referring to common retrieval methods in long-horizon tasks[19, 12], we set up three memory architectures: (1) **Naive Long-Context**. No external tools is used; memories from preceding visits are greedily injected into the LLM’s context window. (2) **RAG**. *text-embedding-3-small*[14] is used to vectorize memory entries; the agent can retrieve Top-K similar memory fragments when answering the question. (3) **Agent Memory System**. A dynamically updated external storage layer is included. When answering questions, the agent can spontaneously search based on the request and recall appropriate memories. We introduce Mem0[1], a product-ready agent memory architecture, in the following experiments.

3.2 Results

Factual QA We evaluate **Naive Long-Context LLMs** using *qwen-turbo*, comparing window sizes $\{3, 5, 7\}$ (*i.e.*, $m \in \{1, 2, 3\}$ in $\mathcal{M}_E$) and full history ($m = \infty$) against the **Agent Memory system** (Mem0). Results in Table 2 report **Lab-T** (recall) and **Lab-F** (rejection) to assess hallucination impact. Table 2 reveals that **naive-LLM’s performance decays severely as history grows**, dropping from 0.882 to 0.423 in explicit retrieval, with low Lab-F scores (≤ 0.570) highlighting a pervasive hallucination bias. **The performance of agent memory system is strongly correlated with the specific task type**; while Mem0 achieves near-optimal results in explicit Lab-T (0.993) and Medication (0.983), its relative performance (Overall 0.331) remains inferior. This gap reveals that: first, agents struggle to generate precise queries for imaging reports even with few-shot prompting; interestingly, when falling back to vector-based search, relative semantic queries outperform explicit ID/timestamp matching, as the latter lacks sufficient embedding density in vector space; second, the agent’s memory architecture lacks a systematic indexing of relationships between events.

Table 2: Performance on Factual Retrieval under Explicit and Relative Settings.

Model	Explicit							Relative						
	L-T ^f	L-F ^g	L-O ^h	Med.	Img.	Adm/Dis	Avg.	L-T ^f	L-F ^g	L-O ^h	Med.	Img.	Avg.	
$m = 1$	<u>0.96</u>	0.57	0.83	<u>0.84</u>	0.89	<u>0.97</u>	0.88	0.91	0.58	0.80	0.78	0.84	0.80	
$m = 2$	0.92	<u>0.51</u>	0.78	0.71	<u>0.81</u>	0.92	<u>0.81</u>	0.86	0.52	<u>0.74</u>	<u>0.64</u>	<u>0.74</u>	<u>0.71</u>	
$m = 3$	0.86	<u>0.50</u>	0.74	0.61	<u>0.75</u>	0.87	<u>0.74</u>	<u>0.80</u>	0.50	<u>0.70</u>	<u>0.55</u>	0.68	<u>0.64</u>	
$m = \infty$	0.63	0.29	0.51	0.28	0.41	0.46	0.42	0.52	0.28	0.44	0.23	0.36	0.34	
Mem0	0.99	0.43	<u>0.80</u>	0.98	0.07	0.98	0.74	0.71	<u>0.57</u>	0.66	0.17	0.13	0.33	

^fL-T: Lab-T, recall of existing lab indicators. ^gL-F: Lab-F, rejection of not found lab indicators. ^hL-O: Overall accuracy of lab events.

Table 3: Temporal reasoning performance (Kendall’s τ) on LongMedBench.

Model	<i>visit_cloze</i>			<i>visit_sorting</i>			<i>joint_sorting</i>		
	N	Mean	Std	N	Mean	Std	N	Mean	Std
deepseek-v3.2	6776	0.316	0.352	765	0.258	0.523	2878	0.132	0.346
gpt-5-mini	6776	<u>0.925</u>	0.220	765	<u>0.376</u>	0.552	2878	<u>0.295</u>	0.439
deepseek-v3.2-thinking	6776	0.969	0.124	765	0.424	0.549	2878	0.330	0.425
qwen-turbo	6776	0.047	0.257	765	0.114	0.456	2878	0.033	0.256

Temporal Reasoning Table 3 presents Kendall’s τ for temporal reasoning across three progressive tasks. In event-level **visit cloze**, top models achieve near-perfect performance (*gpt-5-mini*: 0.925), while weaker models struggle (*qwen-turbo*: 0.046). Enabling thinking mode substantially boosts reasoning (*deepseek-v3.2-thinking*: 0.969 vs. *deepseek-v3.2*: 0.316). This confirms that **strong LLMs can effectively utilize explicit timestamp cues** for event-level ordering, regardless of the option numbers (the length of the event stream), as illustrated in Figure 3a.

Moving to **Visit Sorting**, where timestamps are absent so that models must sort five complete visit summaries based solely on clinical progression, performance drops sharply (best: *deepseek-v3.2-thinking* 0.423). This reveals that **implicit temporal reasoning in visit-level remains challenging**.

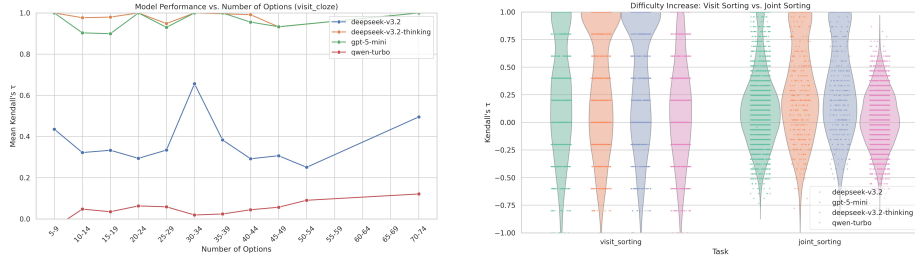
(a) Mean Kendall’s τ vs. number of options in *visit_cloze*.(b) Comparison between *joint_sorting* and *visit_sorting*.

Fig. 3: Temporal reasoning analysis.

The difficulty further escalates in **Joint Sorting**, where each visit summary is split into admission and discharge fragments, requiring simultaneous event-level pairing and visit-level sorting. Performance declines to 0.330, demonstrating the compounding complexity. Figure 3b shows this clear descending trend, confirming that when crucial information is fragmented at both event and visit level, **implicit temporal reasoning in realistic multi-visit scenarios remains a major challenge**.

Table 4: Long-context LLM performance on Long-Horizon Decision Making.

Model	Ctx ⁱ	Note Memory				Event Memory			
		T3-A	T3-D	T3-N	Avg	T3-A	T3-D	T3-N	Avg
qwen-turbo	128K	0.475	0.604	<u>0.309</u>	<u>0.434</u>	<u>0.494</u>	0.604	<u>0.309</u>	0.420
deepseek-v3.2	128K	0.478	<u>0.607</u>	0.278	0.422	0.493	0.607	0.304	<u>0.439</u>
gpt-5-mini	400K	<u>0.478</u>	0.625	0.324	0.445	0.494	0.607	0.335	0.452
deepseek-v3.2-thinking	128K	0.471	0.578	0.294	0.420	0.466	0.593	0.290	0.419

ⁱCtx: the maximum context length of LLM.

Long-Horizon Decision Making We first evaluate the performance of Naive Long-Context. All models are limited to a maximum context of 128K tokens. As shown in Table 4, structured event memory outperforms note memory in most models in decision making. *gpt-5-mini* performs best. Notably, for *deepseek-v3.2*, thinking patterns did not significantly improve decision performance. This is due to thinking mode makes the agent more conservative, attempting to ask question to get instant information rather than referring to past memories.

To analyze the Lost-in-Middle[11] effect, using *qwen-turbo* as the baseline model, we (1) adjusted the number of injected historical visits and (2) implement RAG and Mem0 architecture as retrieval source in separate experiments. As shown in Table 5, although the difference is subtle, close to that under full memory conditions, increasing the context length makes the agent perform worse. The performance differences between Mem0 and RAG are minimal, both close to the baseline performance, demonstrating that existing memory augmentation architectures offer limited gains for long-term decision tasks.

Table 5: Ablation study. Left: visit injection (n). Right: memory architectures.

Visit Injection					Memory Architecture						
n	Memory	T3-A	T3-D	T3-N	Avg	Arch	Memory	T3-A	T3-D	T3-N	Avg
0 (Base) ^j	–	0.49	0.61	0.32	0.45	RAG	event	0.45	0.51	0.29	0.40
2	event	<u>0.49</u>	<u>0.60</u>	0.32	<u>0.44</u>	Mem0	event	<u>0.47</u>	0.62	0.31	0.44
5	event	<u>0.48</u>	<u>0.60</u>	0.31	0.43	RAG	note	0.48	<u>0.60</u>	0.31	<u>0.44</u>
2	note	0.48	<u>0.60</u>	<u>0.32</u>	0.44	Mem0	note	0.46	0.56	0.29	0.41
5	note	0.48	<u>0.60</u>	0.32	0.44						

^jBaseline with no memory content injected.

Therefore, under the current LongMedBench setting, decision quality is not strongly correlated with the amount of retrieved historical information, while **the agent performance depends more on the model’s clinical reasoning capacity within the immediate context.** Accordingly, we need to design tasks that amplify deeper reasoning requirements and higher decision interdependencies in an attempt to better reveal the limitations of current models and more effectively assess their long-term clinical reasoning capabilities.

4 Conclusion

We introduce LongMedBench, a benchmark using real-world EHR data to evaluate medical agents in long-horizon clinical reasoning. By converting MIMIC-IV records into multi-session event streams, we assess agents across three memory types with varying tasks. Experiments reveal that while state-of-the-art models handle explicit timestamps well, implicit temporal reasoning remains a significant bottleneck. As history scales, models increasingly rely on immediate context rather than cross-session integration. Furthermore, while memory augmentation improves retrieval, decision-making performance is highly task-sensitive and remains limited by the model’s instant reasoning capacity. Our results highlight a critical gap in handling time-dependent clinical tasks for medical agents, necessitating future optimization of cross-session memory augmentation architectures.

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